

Satya Sai Vasamsetty

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TECHNICAL SKILLS

Languages	Python, C/C++, SQL, JavaScript, Bash Scripting
ML / CV	PyTorch, OpenCV, YOLO (v5/v8/World), Person Re-ID, Facial Recognition, ONNX, NCNN, Albumentations
Gen AI / LLMs	Qwen VLM, Gemini Multimodal, RAG (Retrieval-Augmented Generation), LangChain, LLM Fine-tuning, Prompt Engineering, Agentic AI, Multimodal AI, Rate Limiting, Load Balancing
Infra & Tools	Docker, AWS, Linux, Git, Nginx, Raspberry Pi, Node.js, React
Databases	MySQL, MongoDB, Redis
Concepts	Edge Deployment, Object Detection & Tracking, Generative AI Pipelines, Vector Embeddings, REST APIs, Networking, OS Fundamentals

EXPERIENCE

Senior Deep Learning Engineer <i>Tarsyer Insights LLP</i>	January 2024 – Present <i>Pune, Maharashtra, India</i>
- Delivered 9 production Computer Vision and Generative AI systems across 1,308+ locations for clients: Trent, Royaloak, Dr Batras, Mahindra Logistics, Ultratech Cement, Oerlikon Balzers, Maruti Suzuki, and ArcelorMittal. - Managed 1,000+ edge devices via a central server with zero-data-loss MySQL pipelines; designed a Git + Python architecture for fleet-wide software versioning and deployment. - Built and self-hosted open-source Generative AI inference (Qwen LLM) with custom rate limiters and load balancers; designed agentic workflows for multi-step reasoning and structured output generation. - Led full project lifecycle — client requirement gathering, model selection, data pipeline design, edge deployment, and production monitoring.	

PROJECTS

Retail Store Production Monitoring <i>YOLO · Person Re-ID · Qwen VLM · Gemini · Generative AI · Facial Recognition · ResNet</i>	<i>Trent Limited (Westside & Zudio)</i>
Deployed across 1,200 stores on 1,000+ edge devices; 10 use-cases — footfall (M/F/child), dwell time, fire-exit blockage, crowd alerts, store manager tracking, intrusion detection, and back-office cleanliness via Generative AI (multimodal LLM).	
Customer Analytics System <i>MobileNet · Object Tracking · Person Re-ID · ResNet · Cosine Distance</i>	<i>Royaloak Furniture</i>
Accurate group and unique customer counting across 86 stores; re-identification eliminates duplicate counts across frames for reliable footfall KPIs.	
Patient & Operations Monitoring <i>YOLO · NCNN · Person Re-ID · Gemini · Generative AI · Audio Analysis</i>	<i>Dr Batras</i>
10 use-cases across 11 clinics — patient flow, dwell time, doctor-patient time, staff uniform compliance, greeting check, audio interaction analysis via Generative AI (Gemini multimodal LLM), and real-time device health alerts.	
Warehouse & Dock Monitoring <i>YOLO · MobileNet · Gemini · Generative AI · YOLO-World · CLIP · Clustering</i>	<i>Mahindra Logistics</i>
7 use-cases across 3 hubs (Pune, Jaipur, Hyderabad) — dock activity, truck number-plate detection, loitering alerts, pallet counting, mobile-phone usage detection via Generative AI (Gemini multimodal), and suspicious-activity monitoring.	
Star Bazaar Store Monitoring <i>YOLO · Binary Classifier · ONNX · ResNet Re-ID · Facial Recognition</i>	<i>Trent Limited</i>
6 use-cases across 3 sites — footfall, shelf empty-box detection for fruits/vegetables/staples (binary classifier, ONNX), fire-exit monitoring, and cash till activity.	
Tool Counting & Measurement <i>YOLO · Albumentations · PCA · Linear Regression · Raspberry Pi</i>	<i>Oerlikon Balzers</i>
Counts 10–600 tools on a metallic tray at 99.98% accuracy; measures diameter (± 0.5 mm) and length (± 1 mm) in <1 s on Raspberry Pi. Reduced manual work by 90%. Trained on 249 images with custom augmentation.	
Warehouse Production Monitoring <i>MobileNet · Object Tracking · Object Detection</i>	<i>Ultratech Cement</i>
Automated cement bag counting (entry & exit) across 2 production sites, eliminating manual tracking and improving inventory accuracy.	
Activity Assessment System <i>Object Detection · Color Segmentation · Motion Detection · Contour Detection · Sequence Logic</i>	<i>Maruti Suzuki</i>
CV system monitoring 10 assembly activities at 99.7% accuracy — tracks task duration, idle time, and step transitions for trainer performance evaluation.	
Crane Hook Safety Detection <i>YOLOX Nano · ONNX Runtime · ZeroMQ · Raspberry Pi · Moxa PLC · RTSP</i>	<i>ArcelorMittal</i>
Safety-critical real-time CV system classifying crane ladle hook states (hooked / mishooked / active) and triggering PLC signals (safe / unsafe / warning) to prevent accidents during molten metal transfers; multi-camera consensus with gated 10-second hold before any signal fires.	

EDUCATION

Jawaharlal Nehru Technological University <i>Bachelor of Technology in Computer Science</i>	Aug 2019 – July 2023 <i>Sangareddy, Telangana, India</i>
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ACTIVITIES & ACHIEVEMENTS

- Volunteered as Python and Web Development mentor at College Technical Fest.
- Conducted an interactive Git & GitHub workshop for students at College Technical Fest.
- CODE ADDICTS — created and hosted a programming contest at YUGA2K22 on HackerRank.